

ASSIGNMENT

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**Project Title: Design And Development of An Intelligent
Emergency Medical Decision-Support Agent Using Search, Logic
and Expert-System Techniques**

ABSTRACT

Emergency departments frequently receive multiple patients with different symptoms, vital signs and medical histories. Efficient prioritization of patients is therefore an important decision-support problem. Artificial Intelligence techniques such as knowledge representation, heuristic search, expert systems, logical inference and intelligent agents can be integrated to support structured decision-making.

This project presents the design and development of an **Intelligent Emergency Medical Decision-Support Agent (IEMDSA)** for an emergency department. The system receives patient symptoms, vital information and medical history and processes them using a combination of Artificial Intelligence techniques. The emergency decision-making process is represented as a state-space problem, and *A Search** is implemented to identify an appropriate assessment path. **Greedy Best-First Search** is also implemented to provide a comparison between search approaches.

Medical knowledge is represented using **propositional logic and First-Order Logic (FOL)**. The system demonstrates **unification and resolution** to illustrate logical reasoning. A rule-based expert system containing **13 production rules** is implemented using **forward chaining and backward chaining**. An intelligent-agent architecture incorporates sensors, knowledge representation, reasoning, search, decision-making, learning and actuators.

The prototype is implemented in **Python using Visual Studio Code** and evaluated using four sample patient scenarios, P1, P2, P3 and P4. The system generates risk scores, priorities, assessment recommendations and resource-allocation suggestions. The results demonstrate how different patient observations lead to different inference results and assessment recommendations.

The proposed system is an **academic decision-support prototype and not a medical diagnostic or treatment system**. The project demonstrates the integration of multiple AI concepts into a single practical application while emphasizing explainability, safety, privacy, ethical use and human oversight.

Keywords: Artificial Intelligence, Expert System, Intelligent Agent, A* Search, Greedy Best-First Search, Propositional Logic, First-Order Logic, Unification, Resolution, Forward Chaining, Backward Chaining, Healthcare Decision Support.

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CHAPTER 1- INTRODUCTION

1.1 Background

Artificial Intelligence is concerned with developing computational systems capable of reasoning, problem solving, learning and making decisions. Search algorithms, logic-based reasoning and expert systems are important areas of AI because they provide structured methods for solving complex problems.

Healthcare emergency departments present a challenging decision-making environment because several patients can arrive simultaneously, each with different symptoms, vital signs and medical histories. Emergency staff must quickly interpret available information and determine which cases require greater attention.

The assignment scenario requires an **Intelligent Emergency Medical Decision-Support Agent (IEMDSA)** capable of assisting doctors and emergency staff with patient prioritization, possible condition identification, action recommendation and resource allocation.

The proposed system integrates search, logic, inference, expert-system techniques, intelligent-agent architecture and learning within a single prototype. The objective is specifically to demonstrate AI integration rather than replace healthcare professionals.

1.2 Problem Statement

The project addresses the following problem:

Design and develop an Intelligent Emergency Medical Decision-Support Agent that can process patient information, reason using logical knowledge, search for an appropriate assessment sequence, prioritize patients and provide explainable recommendations.

The hospital environment contains limited resources such as:

- Emergency beds
- Doctors
- Diagnostic equipment
- Specialist availability

The system therefore needs to support prioritization and resource allocation.

1.3 Motivation

Manual emergency prioritization can involve large amounts of patient information. An AI-based decision-support system can organize information into structured facts and apply predefined reasoning rules.

The motivation for this project is to demonstrate how AI can:

- Represent real-world medical observations.
- Infer risk-related information.
- Search through possible assessment states.
- Apply expert knowledge.
- Compare reasoning strategies.
- Learn from historical cases.
- Provide explainable recommendations.

1.4 Objectives

The main objectives are:

1. To formulate emergency decision-making as a state-space problem.
2. To define initial and goal states.
3. To define actions, operators and path costs.
4. To implement A* Search.
5. To implement Greedy Best-First Search for comparison.
6. To define an appropriate heuristic function.
7. To represent medical knowledge using propositional logic.
8. To represent additional knowledge using First-Order Logic.
9. To demonstrate unification.
10. To demonstrate resolution.
11. To develop a rule-based expert system.

12. To implement forward chaining.
13. To implement backward chaining.
14. To design an intelligent-agent architecture.
15. To demonstrate learning from historical cases.
16. To test the system using four patient scenarios.
17. To validate and compare the implemented approaches.
18. To discuss privacy, ethics, safety, bias and human oversight.

These objectives correspond directly to the functional requirements in the assignment.

1.5 Scope

The scope includes:

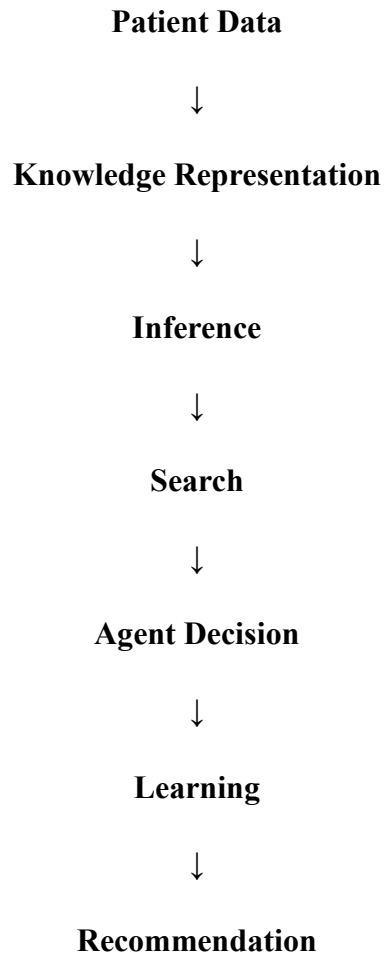
- Emergency patient prioritization.
- Risk classification based on predefined rules.
- Assessment-path selection.
- Logical inference.
- Expert-system reasoning.
- Search-based decision support.
- Learning from historical cases.
- Resource-allocation recommendations.
- Explainable output.

The system is limited to the predefined patient scenarios and knowledge supplied by the student. The assignment explicitly requires recommendations to be treated as **decision-support outputs rather than clinical prescriptions**.

1.6 Significance

The major significance of the project is the integration of multiple AI concepts.

Instead of implementing only one algorithm, the system combines:



This integrated approach directly satisfies the assignment objective of demonstrating search, logic, inference, intelligent agents and expert-system techniques in one computational prototype.

CHAPTER 2 - PROBLEM UNDERSTANDING AND FORMULATION

2.1 Emergency Decision-Support Problem

The hospital receives four sample patients.

Table 2.1: Patient Input Scenarios

Patient	Symptoms / Observations	Vital Information	Medical History
P1	Chest pain, sweating, shortness of breath	High heart rate	Hypertension
P2	High fever, cough, breathing difficulty	Low oxygen saturation	No known cardiac history
P3	Severe headache, one-sided weakness	High blood pressure	Hypertension
P4	Abdominal pain, vomiting, fever	Elevated temperature	No major previous illness

These are the four scenarios specified by the assignment.

2.2 Available Information

The system uses:

- Patient ID
- Symptoms
- Heart rate
- Oxygen saturation
- Temperature
- Blood pressure
- Medical history

This information is converted into logical facts that can be processed by the inference engine.

2.3 Assumptions

The following assumptions are used:

1. Patient observations are assumed to be correctly entered.
2. The predefined knowledge base represents the available expert knowledge.
3. Each production rule has a deterministic result.
4. Search transition costs are predefined.
5. The risk score is a prototype score and is not a clinically validated score.
6. Historical cases are assumed to represent useful examples for the learning component.
7. Recommendations are decision-support outputs.
8. Final medical decisions remain under human supervision.

The assignment specifically requires assumptions to be clearly stated.

2.4 Problem Formulation

Table 2.2: Problem Formulation

Component	Description
Initial State	Patient arrives with symptoms, vital signs and medical history
Goal State	Appropriate assessment and priority decision are generated
States	Patient Arrival, Initial Assessment, Vital Check, Risk Assessment, Assessment, Goal
Actions/Operators	Initial assessment, checking vitals, risk assessment, selecting assessment, generating decision
Path Cost	Cost associated with each state transition
Constraints	Limited resources, predefined knowledge, explainability and safety
Available Knowledge	Patient facts, logical rules, production rules and historical cases

Expected Output	Risk score, priority, assessment recommendation, resource allocation and explanation
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This directly covers the problem-formulation table required for the 10-mark section of the rubric.

2.5 State-Space Representation

The emergency decision problem is represented using the following states:

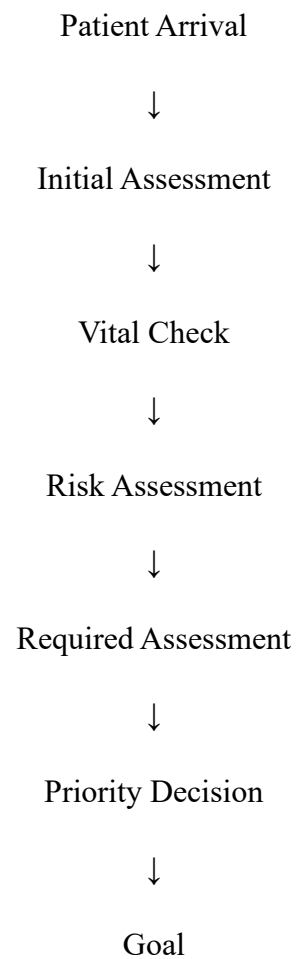


Table 2.3: State-Space Representation

State	Description
S0 – Patient Arrival	Patient information enters the system
S1 – Initial Assessment	Basic patient information is processed
S2 – Vital Check	Vital information is evaluated

S3 – Risk Assessment	Risk-related facts are inferred
S4 – Assessment Selection	Appropriate assessment is selected
S5 – Priority Decision	Priority is generated
S6 – Goal	Final recommendation is produced

The assignment specifically requires the state-space, operators, state transitions and path cost to be defined.

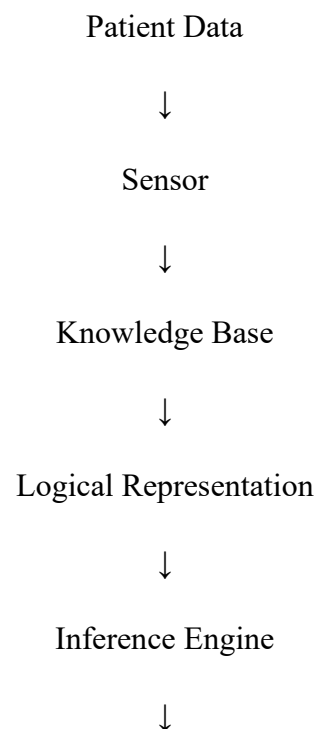
CHAPTER 3 – SYSTEM DESIGN AND AI METHODOLOGY

3.1 Overall System Architecture

The proposed architecture contains:

1. Patient Data/Input Module
2. Sensor Module
3. Knowledge Base
4. Logical Reasoning Module
5. Expert-System Inference Engine
6. A* Search Module
7. Greedy Search Module
8. Agent Decision Module
9. Learning Component
10. Actuator/Recommendation Module

Overall Flow



A* / Greedy Search



Agent Decision



Learning Component



Recommendation

3.2 Propositional Logic

Propositional logic represents patient observations as facts that can be either true or false.

Table 3.1: Propositional Knowledge

No.	Fact / Rule
P1	Chest_Pain(P1)
P2	Sweating(P1)
P3	Shortness_Of_Breath(P1)
P4	High_Heart_Rate(P1)
P5	Hypertension_History(P1)
P6	Breathing_Difficulty(P2)
P7	Cough(P2)
P8	High_Fever(P2)
P9	Low_Oxygen(P2)
P10	Severe_Headache(P3)
P11	One_Sided_Weakness(P3)
P12	High_Blood_Pressure(P3)
P13	Abdominal_Pain(P4)

P14	Vomiting(P4)
P15	Fever(P4)

The assignment requires at least 10 medical facts/rules using propositional or first-order logic.

3.3 First-Order Logic

First-Order Logic provides a more general representation by using predicates, variables and relationships.

Table 3.2: FOL Representation

No.	FOL Representation
1	ChestPain(P1)
2	Sweating(P1)
3	BreathingDifficulty(P2)
4	LowOxygen(P2)
5	SevereHeadache(P3)
6	OneSidedWeakness(P3)
7	HighBloodPressure(P3)
8	AbdominalPain(P4)
9	Vomiting(P4)
10	Fever(P4)

Example general rules:

$$\forall x \text{ ChestPain}(x) \wedge \text{Sweating}(x) \rightarrow \text{CardiacRisk}(x)$$

$$\forall x \text{ BreathingDifficulty}(x) \wedge \text{LowOxygen}(x)$$

$$\rightarrow \text{RespiratoryRisk}(x)$$

$$\forall x \text{ SevereHeadache}(x) \wedge \text{OneSidedWeakness}(x)$$

$$\rightarrow \text{NeurologicalRisk}(x)$$

$\forall x \text{ AbdominalPain}(x) \wedge \text{Vomiting}(x)$

$\rightarrow \text{AbdominalRisk}(x)$

$\forall x \text{ Fever}(x) \rightarrow \text{InfectionRisk}(x)$

3.4 Unification

Unification determines whether two logical expressions can be made identical through substitutions.

Example 1

$\text{HasSymptom}(X, \text{ChestPain})$

$\text{HasSymptom}(P1, \text{ChestPain})$

Substitution:

$X = P1$

Therefore:

$\text{HasSymptom}(P1, \text{ChestPain})$

Example 2

$\text{HasSymptom}(X, \text{SevereHeadache})$

$\text{HasSymptom}(P3, \text{SevereHeadache})$

Substitution:

$X = P3$

Therefore:

$\text{HasSymptom}(P3, \text{SevereHeadache})$

Table 3.3: Unification Examples

Example	Expression 1	Expression 2	Substitution
1	$\text{HasSymptom}(X, \text{ChestPain})$	$\text{HasSymptom}(P1, \text{ChestPain})$	$X = P1$

2	HasSymptom(X, SevereHeadache)	HasSymptom(P3, SevereHeadache)	X = P3
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The assignment explicitly requires two unification examples and the substitutions used.

3.5 Resolution

Resolution is used to demonstrate logical inference by converting statements into clauses and resolving complementary literals.

Knowledge Base

ChestPain(P1)

ChestPain(x) ∧ Sweating(x) → CardiacRisk(x)

CardiacRisk(x) → UrgentAssessment(x)

Query

UrgentAssessment(P1)

The rules are converted into suitable clauses.

The reasoning establishes:

ChestPain(P1)

↓

CardiacRisk(P1)

↓

UrgentAssessment(P1)

The implemented resolution demonstration reaches the empty clause, showing that the query follows from the selected knowledge.

The assignment requires the knowledge base, query, clausal representation, resolution steps and final conclusion.

3.6 Production-Rule Expert System

The prototype contains **13 production rules**, which exceeds the assignment minimum of 10 rules. The assignment explicitly requires a rule-based expert system containing at least 10 production rules.

Table 3.4: Production Rules

Rule	Production Rule
R1	IF chest pain AND sweating THEN cardiac risk
R2	IF cardiac risk AND high heart rate THEN cardiac warning
R3	IF cardiac risk AND shortness of breath THEN very high risk
R4	IF high fever AND cough THEN respiratory risk
R5	IF breathing difficulty AND low oxygen THEN respiratory risk
R6	IF severe headache AND one-sided weakness THEN neurological risk
R7	IF neurological risk AND high blood pressure THEN high priority
R8	IF abdominal pain AND vomiting THEN abdominal risk
R9	IF abdominal risk AND fever THEN infection risk
R10	IF infection risk THEN medical assessment required
R11	IF very high risk THEN immediate attention
R12	IF respiratory risk THEN urgent respiratory assessment
R13	IF neurological risk THEN urgent neurological assessment

3.7 Forward Chaining

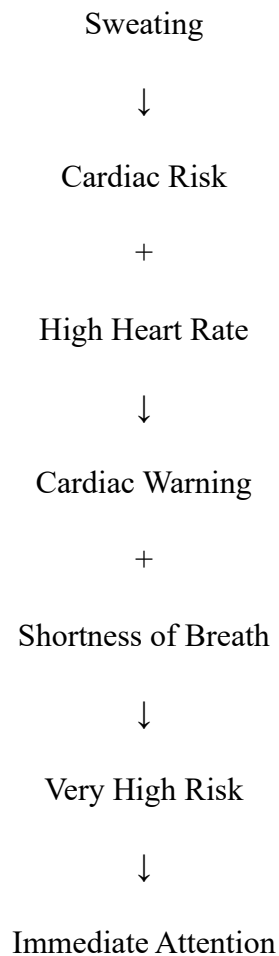
Forward chaining is a **data-driven reasoning technique**.

The process begins with known patient facts and applies rules whose conditions are satisfied.

Example: P1

Chest Pain

+



The actual execution shows the rules fired for P1 as:

R1, R2, R3, R11

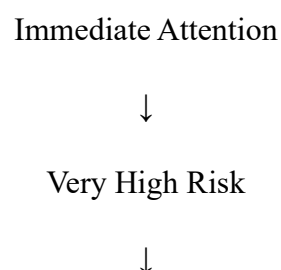
3.8 Backward Chaining

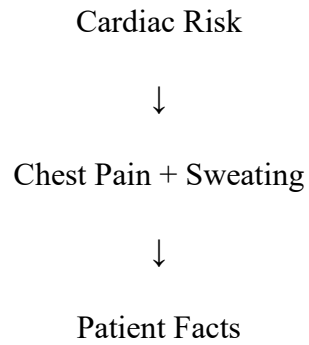
Backward chaining is a **goal-driven reasoning technique**.

The system begins with a target conclusion and determines whether the available facts and rules can support that conclusion.

Example:

Goal:





The prototype verifies the required goal using the knowledge generated during reasoning.

The assignment specifically requires comparison of forward and backward chaining in terms of direction, rule exploration, suitability, explainability and computational requirements.

3.9 A* Search

A* Search is used to select an assessment path.

The evaluation function is:

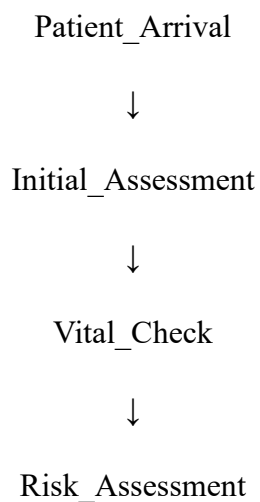
$$f(n) = g(n) + h(n)$$

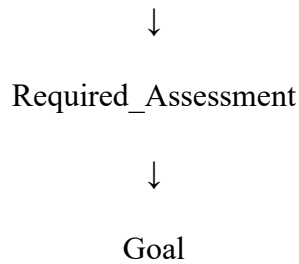
where:

- $g(n)$ = actual cost from the initial state to node n .
- $h(n)$ = estimated cost from node n to the goal.
- $f(n)$ = estimated total cost.

The assignment explicitly provides this A* formulation.

State Sequence





The implemented execution shows:

Path Cost = 5

States Explored = 6

3.10 Greedy Best-First Search

Greedy Best-First Search selects the state with the most promising heuristic value.

It primarily considers:

$$f(n) = h(n)$$

Unlike A*, it does not combine the accumulated path cost and heuristic in the same way.

The prototype implements Greedy Best-First Search to provide an alternative search approach for comparison.

3.11 Intelligent Agent Architecture

The proposed agent is designed as a **Learning Agent**.

Table 3.5: Intelligent Agent Components

Component	Description
Environment	Hospital emergency department
Sensors	Patient symptoms and vital observations
Percepts	Symptoms, heart rate, oxygen, temperature, BP, history
Knowledge Base	Facts, rules and historical information
Inference Engine	Forward/backward chaining
Search Engine	A* and Greedy Search

Decision Module	Priority and assessment selection
Learning Component	Historical-case-based priority estimation
Actuators	Recommendation and resource allocation
Goal	Support emergency prioritization and assessment

3.12 Environment Classification

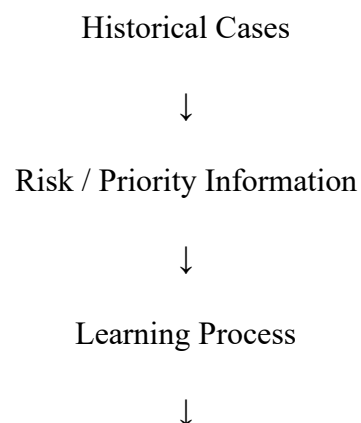
Table 3.6: Environment Classification

Property	Classification	Explanation
Observability	Partially observable	Complete patient information may not always be available
Determinism	Partially deterministic	Patient information can change
Episodic/Sequential	Sequential	Decisions influence subsequent assessment
Static/Dynamic	Dynamic	Patient condition may change
Discrete/Continuous	Mixed	Rules are discrete while vital measurements may be continuous
Number of agents	Multi-agent	Doctors, nurses and AI system interact

3.13 Learning Component

The learning component uses historical patient cases.

The workflow is:



Updated Knowledge



New Patient



Improved Priority Estimation

The implementation uses historical scores such as critical, high, medium and low cases to provide a learning-based priority estimate.

The assignment requires the learning sequence:

Input → Learning Process → Updated Knowledge/Model → Improved Decision.

CHAPTER 4 - IMPLEMENTATION

4.1 Development Environment

Table 4.1: Development Environment

Technology	Purpose
Python	Main programming language
Visual Studio Code	Development environment
Python Terminal	Program execution
Production Rules	Expert-system reasoning
A* Search	Heuristic search
Greedy Search	Search comparison
Propositional Logic	Fact representation
First-Order Logic	General logical representation
Forward Chaining	Data-driven inference
Backward Chaining	Goal-driven inference
Historical Cases	Learning component

The assignment permits Python, Jupyter Notebook, Prolog, CLIPS or another suitable AI/expert-system environment.

4.2 Implementation Workflow

The implemented system follows:

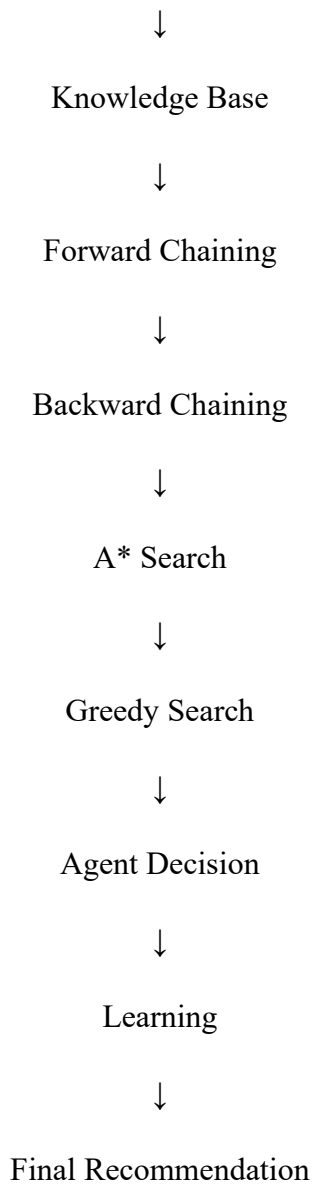
Patient Data



Sensor



Patient Facts



This corresponds directly to the required prototype flow:

Patient Data → Knowledge Base → Search/Inference → Agent Decision → Recommendation.

4.3 Patient P1 Implementation

P1 contains:

- Chest pain
- Sweating
- Shortness of breath
- High heart rate

- Hypertension history

The propositional facts generated by the system include:

chest_pain

high_heart_rate

hypertension_history

shortness_of_breath

sweating

The forward-chaining engine fires:

R1, R2, R3 and R11

The resulting facts include cardiac risk, cardiac warning, very high risk and immediate attention.

The system produces:

- **Risk Score:** 11
- **Rule-Based Priority:** CRITICAL
- **Learning-Based Priority:** CRITICAL
- **Assessment:** Urgent assessment
- **Resource:** Emergency team + continuous monitoring

```
=====
PROCESSING PATIENT: P1
=====

[SENSOR] Receiving patient data: P1

Patient Information:
Symptoms: chest_pain, sweating, shortness_of_breath
Heart Rate: high
Oxygen: normal
Temperature: normal
Blood Pressure: not recorded
History: ['hypertension']

Propositional Facts:
['chest_pain', 'high_heart_rate', 'hypertension_history', 'shortness_of_breath', 'sweating']

Forward Chaining:
Initial Facts: ['chest_pain', 'high_heart_rate', 'hypertension_history', 'shortness_of_breath', 'sweating']
Fired Rules: ['R1', 'R2', 'R3', 'R11']
Inferred Facts: ['cardiac_risk', 'cardiac_warning', 'chest_pain', 'high_heart_rate', 'hypertension_history', 'immediate_attention', 'shortness_of_breath', 'sweating', 'very_high_risk']

Backward Chaining:
Goal: immediate_attention
Result: True
  FACT: immediate_attention
```

Figure 4.6: P1 Processing – Forward and Backward Chaining

```

A* SEARCH
-----
Target Assessment: send_to_urgent_assessment
Path:
  Patient_Arrival --[perform_initial_assessment]--> Initial_Assessment
  Initial_Assessment --[check_vitals]--> Vital_Check
  Vital_Check --[perform_risk_assessment]--> Risk_Assessment
  Risk_Assessment --[send_to_urgent_assessment]--> send_to_urgent_assessment
  send_to_urgent_assessment --[generate_priority_decision]--> Goal
  Goal
Path Cost: 5
States Explored: 6

GREEDY BEST-FIRST SEARCH
-----
Target Assessment: send_to_urgent_assessment
Path Cost: 5
States Explored: 6

AGENT DECISION
-----
Risk Score: 11
Rule-Based Priority: CRITICAL
Learning-Based Priority: CRITICAL
Recommended Assessment: send_to_urgent_assessment
Resource Allocation: Emergency team + continuous monitoring
[ACTUATOR] Patient: P1
[ACTUATOR] Suggested assessment: send_to_urgent_assessment
[ACTUATOR] Priority: CRITICAL

```

Figure 4.7: P1 A* Search and Agent Decision

4.4 Patient P2 Implementation

P2 contains:

- High fever
- Cough
- Breathing difficulty
- Low oxygen saturation

The generated facts include:

breathing_difficulty

cough

high_fever

low_oxygen

The forward-chaining engine fires:

R4, R5 and R12

The system identifies respiratory risk and generates an urgent respiratory assessment.

Output

- **Risk Score:** 10
- **Rule-Based Priority:** CRITICAL
- **Learning-Based Priority:** CRITICAL
- **Assessment:** Respiratory assessment
- **Resource:** Emergency team + continuous monitoring

```
=====
PROCESSING PATIENT: P2
=====

[SENSOR] Receiving patient data: P2

Patient Information:
Symptoms: high_fever, cough, breathing_difficulty
Heart Rate: normal
Oxygen: low
Temperature: high
Blood Pressure: not recorded
History: []

Propositional Facts:
['breathing_difficulty', 'cough', 'high_fever', 'low_oxygen']

Forward Chaining:
Initial Facts: ['breathing_difficulty', 'cough', 'high_fever', 'low_oxygen']
Fired Rules: ['R4', 'R5', 'R12']
Inferred Facts: ['breathing_difficulty', 'cough', 'high_fever', 'high_priority', 'low_oxygen', 'respiratory_risk', 'urgent_respiratory_assessment']

Backward Chaining:
Goal: high_priority
Result: True
FACT: high_priority
```

Figure 4.8: P2 Processing – Forward and Backward Chaining

```
A* SEARCH
-----
Target Assessment: send_to_respiratory_assessment
Path:
Patient Arrival --[perform_initial_assessment]--> Initial_Assessment
Initial_Assessment --[check_vitals]--> Vital_Check
Vital_Check --[perform_risk_assessment]--> Risk_Assessment
Risk_Assessment --[send_to_respiratory_assessment]--> send_to_respiratory_assessment
send_to_respiratory_assessment --[generate_priority_decision]--> Goal
Goal
Path Cost: 5
States Explored: 6

GREEDY BEST-FIRST SEARCH
-----
Target Assessment: send_to_respiratory_assessment
Path Cost: 5
States Explored: 6

AGENT DECISION
-----
Risk Score: 10
Rule-Based Priority: CRITICAL
Learning-Based Priority: CRITICAL
Recommended Assessment: send_to_respiratory_assessment
Resource Allocation: Emergency team + continuous monitoring
[ACTUATOR] Patient: P2
[ACTUATOR] Suggested assessment: send_to_respiratory_assessment
[ACTUATOR] Priority: CRITICAL
```

Figure 4.9: P2 A* Search and Agent Decision

4.5 Patient P3 Implementation

P3 contains:

- Severe headache

- One-sided weakness
- High blood pressure
- Hypertension history

Generated facts include:

high_blood_pressure

hypertension_history

one_sided_weakness

severe_headache

The system fires:

R6, R7 and R13

The inferred information includes:

- Neurological risk
- High priority
- Urgent neurological assessment

Output

- **Risk Score:** 13
- **Rule-Based Priority:** CRITICAL
- **Learning-Based Priority:** CRITICAL
- **Assessment:** Neurological assessment
- **Resource:** Emergency team + continuous monitoring

```

=====
PROCESSING PATIENT: P3
=====

[SENSOR] Receiving patient data: P3

Patient Information:
Symptoms: severe_headache, one_sided_weakness
Heart Rate: normal
Oxygen: normal
Temperature: normal
Blood Pressure: high
History: ['hypertension']

Propositional Facts:
['high_blood_pressure', 'hypertension_history', 'one_sided_weakness', 'severe_headache']

Forward Chaining:
Initial Facts: ['high_blood_pressure', 'hypertension_history', 'one_sided_weakness', 'severe_headache']
Fired Rules: ['R6', 'R7', 'R13']
Inferred Facts: ['high_blood_pressure', 'high_priority', 'hypertension_history', 'neurological_risk', 'one_sided_weakness', 'severe_headache', 'u
t_neurological_assessment']

Backward Chaining:
Goal: high_priority
Result: True
FACT: high_priority

```

Figure 4.10: P3 Processing – Forward and Backward Chaining

```

A* SEARCH
-----
Target Assessment: send_to_neurological_assessment
Path:
  Patient Arrival --[perform_initial_assessment]--> Initial_Assessment
  Initial_Assessment --[check_vitals]--> Vital_Check
  Vital_Check --[perform_risk_assessment]--> Risk_Assessment
  Risk_Assessment --[send_to_neurological_assessment]--> send_to_neurological_assessment
  send_to_neurological_assessment --[generate_priority_decision]--> Goal
  Goal
Path Cost: 5
States Explored: 6

GREEDY BEST-FIRST SEARCH
-----
Target Assessment: send_to_neurological_assessment
Path Cost: 5
States Explored: 6

AGENT DECISION
-----
Risk Score: 13
Rule-Based Priority: CRITICAL
Learning-Based Priority: CRITICAL
Recommended Assessment: send_to_neurological_assessment
Resource Allocation: Emergency team + continuous monitoring
[ACTUATOR] Patient: P3
[ACTUATOR] Suggested assessment: send_to_neurological_assessment
[ACTUATOR] Priority: CRITICAL

```

Figure 4.11: P3 A* Search and Agent Decision

4.6 Patient P4 Implementation

P4 contains:

- Abdominal pain
- Vomiting
- Fever
- High temperature

The generated facts include:

abdominal_pain

fever

high_fever

vomiting

The system fires:

R8, R9 and R10

The resulting facts include:

- Abdominal risk
- Infection risk
- Medical assessment required

Output

- **Risk Score: 5**
- **Rule-Based Priority: MEDIUM**
- **Assessment: General medical assessment**

```
=====
PROCESSING PATIENT: P4
=====

[SENSOR] Receiving patient data: P4

Patient Information:
Symptoms: abdominal_pain, vomiting, fever
Heart Rate: normal
Oxygen: normal
Temperature: high
Blood Pressure: not recorded
History: []

Propositional Facts:
['abdominal_pain', 'fever', 'high_fever', 'vomiting']

Forward Chaining:
Initial Facts: ['abdominal_pain', 'fever', 'high_fever', 'vomiting']
Fired Rules: ['R8', 'R9', 'R10']
Inferred Facts: ['abdominal_pain', 'abdominal_risk', 'fever', 'high_fever', 'infection_risk', 'medical_assessment_required', 'vomiting']

Backward Chaining:
Goal: medical_assessment_required
Result: True
FACT: medical_assessment_required
```

Figure 4.12: P4 Processing – Forward and Backward Chaining


```

A* SEARCH
-----
Target Assessment: send_to_general_assessment
Path:
  Patient Arrival --[perform_initial_assessment]--> Initial_Assessment
  Initial_Assessment --[check_vitals]--> Vital_Check
  Vital_Check --[perform_risk_assessment]--> Risk_Assessment
  Risk_Assessment --[send_to_general_assessment]--> send_to_general_assessment
  send_to_general_assessment --[generate_priority_decision]--> Goal
  Goal
Path Cost: 5
States Explored: 6

GREEDY BEST-FIRST SEARCH
-----
Target Assessment: send_to_general_assessment
Path Cost: 5
States Explored: 6

AGENT DECISION
-----
Risk Score: 5
Rule-Based Priority: MEDIUM
Learning-Based Priority: MEDIUM
Recommended Assessment: send_to_general_assessment
Resource Allocation: Standard medical assessment
[ACTUATOR] Patient: P4
[ACTUATOR] Suggested assessment: send_to_general_assessment
[ACTUATOR] Priority: MEDIUM

```

Figure 4.13: P4 A* Search and Agent Decision

4.7 Knowledge Base Implementation

The implementation contains:

- **13 production rules**
- **15 FOL facts**
- **4 FOL rules**
- **4 patient scenarios**

```

=====
KNOWLEDGE BASE SUMMARY
=====
Production Rules: 13
FOL Facts: 15
FOL Rules: 4
Patients Tested: 4

Production Rules:
R1: chest_pain -> cardiac_risk
R2: chest_pain AND sweating -> cardiac_warning
R3: cardiac_risk AND shortness_of_breath -> very_high_risk
R4: high_fever AND low_oxygen -> respiratory_risk
R5: breathing_difficulty AND low_oxygen -> urgent_respiratory_assessment
R6: one_sided_weakness AND severe_headache -> neurological_risk
R7: high_blood_pressure AND neurological_risk -> urgent_neurological_assessment
R8: abdominal_pain AND vomiting -> abdominal_risk
R9: abdominal_risk AND fever -> infection_risk
R10: infection_risk -> medical_assessment_required
R11: very_high_risk -> immediate_attention
R12: urgent_respiratory_assessment -> high_priority
R13: urgent_neurological_assessment -> high_priority

```

Figure 4.2: Knowledge Base Summary and Production Rules

4.8 Unification Implementation

```

=====
UNIFICATION EXAMPLES
=====
Example 1:
  has_symptom(X, chest_pain)
  has_symptom(P1, chest_pain)
  Substitution: {'X': 'P1'}

Example 2:
  possible_condition(X, cardiac_risk)
  possible_condition(P1, cardiac_risk)
  Substitution: {'X': 'P1'}

```

Figure 4.3: Unification Examples

4.9 Resolution Implementation

```
=====
RESOLUTION
=====
Knowledge Base:
1. chest_pain
2. chest_pain -> cardiac_risk
3. cardiac_risk -> urgent_assessment

Query: urgent_assessment

Clausal form:
C1 = {chest_pain}
C2 = {NOT_chest_pain, cardiac_risk}
C3 = {NOT_cardiac_risk, urgent_assessment}
Negated Query:
C4 = {NOT_urgent_assessment}

Resolution steps:
C1 + C2 -> {'cardiac_risk'}
R1 + C3 -> {'urgent_assessment'}
R2 + C4 -> set()

Conclusion: Empty clause obtained.
Therefore urgent_assessment is logically supported.
=====
```

Figure 4.4: Resolution Demonstration

4.10 Learning Implementation

```
=====
LEARNING COMPONENT
=====
Historical risk score: 13 -> CRITICAL
Historical risk score: 11 -> CRITICAL
Historical risk score: 9 -> HIGH
Historical risk score: 6 -> MEDIUM
Historical risk score: 5 -> MEDIUM
Historical risk score: 2 -> LOW
=====
```

Figure 4.5: Learning Component

4.11 Intelligent Agent Implementation



Figure 4.1: Intelligent Agent Architecture

CHAPTER 5 - RESULTS, TESTING AND VALIDATION

The assignment requires testing with at least four patient scenarios and a results table containing patient input facts, search result, inference result, expert-system decision and final priority.

5.1 Overall Testing Results

Table 5.1: Patient Testing Results

Patient	Risk Score	Search Result	Inference Result	Expert-System Decision	Final Priority
P1	11	Urgent assessment	Cardiac/very high risk	Immediate attention	CRITICAL
P2	10	Respiratory assessment	Respiratory risk	Urgent respiratory assessment	CRITICAL
P3	13	Neurological assessment	Neurological risk/high priority	Urgent neurological assessment	CRITICAL
P4	5	General assessment	Abdominal/infection risk	Medical assessment required	MEDIUM

5.2 P1 Analysis

P1 has multiple cardiac-related observations. The system activates four relevant production rules.

The highest-level inference is:

Chest Pain + Sweating



Cardiac Risk



Very High Risk



Immediate Attention

The final prototype priority is **CRITICAL**.

5.3 P2 Analysis

P2 contains respiratory-related observations.

The reasoning is:

Breathing Difficulty + Low Oxygen



Respiratory Risk



Urgent Respiratory Assessment



CRITICAL Priority

The final risk score is **10**.

5.4 P3 Analysis

P3 produces the highest risk score in the test cases.

Severe Headache

+

One-Sided Weakness



Neurological Risk

+

High Blood Pressure



High Priority



Urgent Neurological Assessment

The final risk score is **13**, resulting in **CRITICAL** priority.

5.5 P4 Analysis

P4 produces a lower risk score compared with the other cases.

Abdominal Pain + Vomiting



Abdominal Risk

+

Fever



Infection Risk



Medical Assessment Required

The final risk score is **5**, resulting in **MEDIUM** priority.

5.6 A* and Greedy Search Comparison

Table 5.2: Search Algorithm Comparison

Patient	A* Cost	A* States	Greedy Cost	Greedy States
P1	5	6	5	6
P2	5	6	5	6
P3	5	6	5	6
P4	5	6	5	6

```

=====
SEARCH ALGORITHM COMPARISON
=====
Patient    A* Path Cost    Greedy Path Cost
-----
P1         5              5
P2         5              5
P3         5              5
P4         5              5

A* Evaluation Function:
f(n) = g(n) + h(n)

g(n): Cost from the initial state
h(n): Estimated cost from current state to goal
f(n): Total estimated path cost

Greedy Best-First Search uses:
f(n) = h(n)

```

Figure 4.14: A* and Greedy Best-First Search Comparison

Analysis

In this implementation, both algorithms produce the same path cost because the prototype uses a simple assessment state space with uniform transition costs.

A* is theoretically more suitable when path costs matter because it combines:

$$g(n) + h(n)$$

Greedy Best-First Search primarily considers the estimated distance to the goal.

Therefore, A* provides a stronger basis for optimal-cost search, while Greedy Best-First Search can be useful when rapid goal-directed exploration is preferred.

The assignment specifically requires the selected search solution to be explained and justified.

5.7 Forward vs Backward Chaining

Table 5.3: Forward and Backward Chaining Comparison

Parameter	Forward Chaining	Backward Chaining
Reasoning direction	Facts → Conclusions	Goal → Supporting facts
Approach	Data-driven	Goal-driven
Starting point	Known patient facts	Desired conclusion

Main purpose	Derive conclusions	Verify a target conclusion
Exploration	Can explore multiple applicable rules	Focuses on rules relevant to the goal
Explainability	High	High
Suitable for	Situations with many available facts	Specific hypotheses/goals
Example	Patient facts → cardiac risk	Immediate attention → supporting facts

5.8 Validation of Reasoning

The reasoning process was validated through:

Validation 1 – Fact Generation

Patient symptoms and observations were successfully converted into propositional facts.

Validation 2 – Rule Activation

Relevant production rules were activated according to patient facts.

Validation 3 – Inference

The inference engine generated new facts from the rule conditions.

Validation 4 – Search

A* Search successfully generated assessment paths.

Validation 5 – Agent Decision

The agent generated priority and assessment recommendations.

Validation 6 – Learning

Historical information was used to generate an additional learning-based priority.

5.9 Final Validation Evidence


```

=====
FINAL VALIDATION / TESTING RESULTS
=====
Patient   Risk Score   Search           Inference Decision           Priority
-----
P1        11           A*               send_to_urgent_assessment     CRITICAL
P2        10           A*               send_to_respiratory_assessment CRITICAL
P3        13           A*               send_to_neurological_assessment CRITICAL
P4         5           A*               send_to_general_assessment     MEDIUM

=====
LEARNING RECORD
=====
Patient: P1 -> Learned Priority: CRITICAL
Patient: P2 -> Learned Priority: CRITICAL
Patient: P3 -> Learned Priority: CRITICAL
Patient: P4 -> Learned Priority: MEDIUM

=====
FINAL SYSTEM SUMMARY
=====
Patients processed: 4
A* Search: Completed
Greedy Best-First Search: Completed
Forward Chaining: Completed
Backward Chaining: Completed
FOL Representation: Completed
Unification: Completed
Resolution: Completed
Expert System: Completed
Intelligent Agent: Completed
Learning Component: Completed
Validation Testing: Completed

=====
PROGRAM EXECUTION COMPLETED
=====

```

Figure 4.15: Final Validation, Testing Results and Learning Records

5.10 Validation Table

Table 5.4: Validation of System Components

Requirement	Implementation	Result
State-space	Patient assessment states	Completed
A* Search	Assessment-path search	Completed
Greedy Search	Search comparison	Completed
Propositional Logic	Patient facts	Completed
FOL	Predicates and rules	Completed
Unification	Two examples	Completed
Resolution	Logical query demonstration	Completed
Production Rules	13 rules	Completed
Forward Chaining	Patient inference	Completed
Backward Chaining	Goal verification	Completed

Agent Architecture	Learning-agent structure	Completed
Learning	Historical cases	Completed
Four scenarios	P1–P4	Completed
Final recommendation	Priority + assessment	Completed

CHAPTER 6 - ANALYSIS, COMPARISON AND JUSTIFICATION

6.1 Comparison of AI Approaches

Table 6.1: Alternative Approaches

Approach	Advantages	Limitations
Rule-Based Expert System	Explainable and easy to trace	Requires manual rule creation
A* Search	Uses cost and heuristic	Requires suitable heuristic
Greedy Search	Simple and goal-directed	May not minimize path cost
Pure Machine Learning	Can learn patterns	Less transparent without explanation mechanisms
Learning Agent	Can improve from historical cases	Depends on quality of historical data
Integrated Approach	Combines reasoning, search and learning	More complex to implement

6.2 Advantages of Proposed System

6.2.1 Explainability

The system displays:

- Patient facts
- Fired rules
- Inferred facts
- Search path
- Risk score
- Priority
- Recommended assessment

Therefore, the reasoning process is traceable.

6.2.2 Integration

The project integrates several AI concepts:



This is one of the strongest aspects of the project because the assignment is specifically designed as an integrated AI decision-support problem.

6.2.3 Structured Decision Making

The state-space model provides a structured sequence for processing a patient.

6.2.4 Multiple Reasoning Strategies

The implementation supports both forward and backward chaining.

6.2.5 Search Comparison

A* and Greedy Best-First Search provide two different approaches for assessment-path selection.

6.2.6 Learning

Historical cases provide an additional mechanism for priority estimation.

6.3 Limitations

Table 6.2: Limitations

Limitation	Explanation
Small dataset	Only four primary patient scenarios are tested
Manual knowledge	Rules are manually defined
Simplified search	State space is relatively small
Uniform costs	Search comparison produces equal path costs
Prototype risk score	Not a clinically validated score
Historical cases	Limited examples for learning
No real-time devices	No direct medical-device integration
No real hospital database	Uses predefined information
Changing patient conditions	Dynamic changes are simplified

The assignment explicitly requires limitations of rule-based and learning-based approaches to be identified.

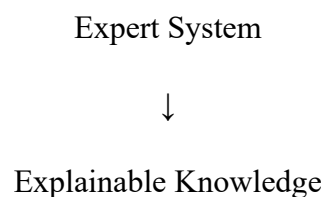
6.4 Why the Integrated Approach Was Selected

A rule-based system provides strong explainability but requires manually created rules.

A learning-based system can identify patterns from historical cases but depends heavily on the quality and representativeness of training data.

A search algorithm provides a structured mechanism for selecting an assessment path but requires an appropriate state representation and heuristic.

Therefore, combining the approaches provides a balanced architecture:



A* Search



Goal-Directed Planning

Learning



Historical Experience

Intelligent Agent



Integration and Decision

This makes the integrated architecture appropriate for the academic objective of the assignment.

CHAPTER 7 - BROADER CONSIDERATIONS AND PROFESSIONAL RESPONSIBILITY

The assignment specifically requires discussion of patient privacy, ethics, safety, explainability, bias, human oversight, accessibility and social impact.

7.1 Patient Data Privacy

Patient information is sensitive and should be protected.

A real system should use:

- Secure storage
- Authentication
- Authorization
- Encryption
- Access control
- Data minimization

The prototype uses predefined academic patient scenarios rather than real identifiable medical records.

7.2 Ethical Use of AI

AI should support healthcare professionals rather than replace them.

The system should not make autonomous medical decisions. Its outputs should be treated as recommendations that require professional review.

7.3 Safety and Reliability

Incorrect patient data can result in incorrect AI outputs.

Therefore:

- Input validation is required.
- Knowledge rules should be carefully verified.
- The system should be tested extensively.

- Human supervision should remain available.

7.4 Explainability

Explainability is a major advantage of the proposed rule-based system.

The system can provide information about:

- Initial facts
- Rules fired
- Inferred facts
- Search path
- Risk score
- Final priority
- Recommended assessment

This allows a user to understand how the recommendation was generated.

7.5 Bias

Historical patient cases may contain bias.

If a learning component uses incomplete or biased historical data, its predictions may also become biased.

Therefore, future systems should use representative and carefully validated datasets.

7.6 Human Oversight

The AI agent should operate under human supervision.

The final decision should be made by qualified healthcare professionals.

This is especially important because the assignment explicitly defines the project as a decision-support prototype rather than a replacement for medical professionals.

7.7 Accessibility

The system should present information clearly and quickly because emergency environments are time-sensitive.

A future graphical interface could use:

- Clear priority indicators
- Patient summaries
- Search-path visualization
- Rule explanations
- Resource status

7.8 Social Impact

AI-based emergency decision support could help organize patient information and assist with resource prioritization.

However, over-reliance on AI could introduce risks if users treat automated recommendations as unquestionable decisions.

Responsible human-AI interaction is therefore necessary.

CHAPTER 8 - CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The project successfully designed and implemented an **Intelligent Emergency Medical Decision-Support Agent** using multiple Artificial Intelligence techniques.

The emergency decision problem was formulated as a state-space search problem with initial state, goal state, intermediate states, actions and path costs. A* Search was implemented using the heuristic evaluation:

$$f(n) = g(n) + h(n)$$

Greedy Best-First Search was also implemented to compare alternative search strategies.

Medical information was represented using propositional logic and First-Order Logic. The project demonstrated unification and resolution to illustrate logical reasoning. A production-rule expert system containing **13 rules** was developed and used with forward and backward chaining.

The intelligent-agent architecture integrates patient perception, knowledge representation, inference, search, decision-making, learning and action. Historical cases are used by the learning component to provide an additional priority estimation.

Four patient scenarios were tested. P1 received a risk score of 11 and critical priority, P2 received a score of 10 and critical priority, P3 received the highest score of 13 and critical priority, while P4 received a score of 5 and medium priority. The assessment recommendations were different for each scenario, demonstrating the influence of patient-specific facts and rules.

The project demonstrates that combining **search, logic, expert systems, intelligent agents and learning** can produce an explainable AI-based decision-support prototype.

However, the system has limitations including a small dataset, manually defined knowledge, simplified search costs and a prototype risk-scoring mechanism. Therefore, it should be considered an academic demonstration and not a clinical diagnostic or treatment system.

8.2 Future Scope

1. Larger Dataset

Future versions can use a larger and more diverse dataset.

2. Advanced Machine Learning

Machine-learning models could be integrated with the existing expert system.

3. Dynamic State Space

The state space could dynamically change according to new patient observations.

4. Real-Time Monitoring

The system could be connected to medical monitoring devices.

5. Advanced Resource Allocation

Future systems could optimize beds, doctors, diagnostic equipment and specialists simultaneously.

6. Improved Explainability

A graphical explanation system could show the complete reasoning chain.

7. Uncertainty Handling

Fuzzy logic or probabilistic reasoning could handle uncertain observations.

8. Human-in-the-Loop Decision Making

Doctors could approve, reject or modify AI recommendations.

CHAPTER 9 - STUDENT REFLECTION

9.1 What did you learn from this assignment beyond the concepts directly taught in class?

This assignment helped me understand how multiple Artificial Intelligence techniques can be integrated into a single practical application. I learned how real-world observations can be converted into logical facts and processed using production rules. I also gained practical knowledge of implementing search algorithms, inference mechanisms, intelligent-agent components and a learning component in Python.

9.2 Which AI technique was most suitable for the proposed problem and why?

The combination of a **rule-based expert system and heuristic search** was most suitable for this prototype. The rule-based system provides explainable decisions because the activated rules can be identified. A* Search provides a structured method for selecting an assessment path. The learning component provides additional support using historical cases.

9.3 What difficulties did you encounter while integrating search, logic and expert-system techniques?

The major difficulty was integrating different AI techniques into one workflow. Each technique uses a different representation and reasoning method. Converting patient observations into facts, applying rules, connecting inference results with search and then passing those results to the intelligent agent required careful organization.

9.4 If you were given additional time and resources, what would you improve?

I would improve the system by using a larger dataset, dynamic patient states and more realistic search costs. I would also develop a graphical user interface, integrate real-time monitoring data and improve the learning model. In addition, I would provide more detailed explanations of the reasoning process and introduce uncertainty handling.

These four reflection questions are explicitly required by the assignment

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